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22 May 1987
SUPERSEDING
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6 December 1986

MILITARY SPECIFICATION

ELECTRICAL DISCHARGE MACHINE, VERTICAL, RAM AND QUILL TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electrical discharge machine having a vertical ram and quill, electro-hydraulic feed, external power supply, complete with equipment and accessories as specified herein.

1.2 Classification. This specification covers electrical discharge machines of the following classes and sizes. The classes and sizes required for specific procurements shall be specified (see 6.2.1).

Class I - Ram head

Size	Ram stroke inches (Min)	Table size, inches (Min) length by width
1	8	20 by 9
2	8	20 by 12
3	12	20 by 12
4	12	36 by 12
5	12	40 by 24

Class II- Quill head

Size	Quill stroke inches (min)	Table size, inches (min) length by width
1	8	26 by 14
2	8	28 by 14

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD-adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1	- Motors and Generators.
ANSI/NEMA ICS 1	- Industrial Control and Systems.
ANSI/NFPA 70	- National Electrical Code.
ANSI/NFPA 79	- Electrical Standard for Industrial Machinery.
ANSI B5.1	- T-Slots, Their Bolts, Nuts and Tongues.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3351.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02	- Manual for Machine Tool Gearing.
AGMA 390.03	- Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54	- Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

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(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3351.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be one of the manufacturer's current models, new and unused. The machine shall be capable of machining through holes, blind holes, shaped cavities, surface contours and trepanning of hardened steels, exotic metals and high-temperature alloys to the tolerances and finishes specified herein. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376, Preferred Metric Units for General Use by the Federal Government.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall

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be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. Unless otherwise specified (see 6.2.1), a digital readout system shall be provided on the electrical discharge machine axis of travel. The digital readout system shall consist of a position feedback device, an electronic display for each axis and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall be capable of providing a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be 0.0002 inch per foot and the repeatability shall be 0.0001 inch. Special requirements for digital readout system components shall be as specified (see 6.2.1). The readout system shall be capable of indicating axis location in both the US and SI systems of measurement.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Provisions shall be made to prevent damage to the feedback device in the event of overtravel.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual digits not less than 0.70 inch high. The console shall be capable of displaying at least six digits simultaneously. The display console shall also provide keyboard entry, memory, absolute/incremental counting, data preset, power-failure indicator, inch or metric conversion, absolute-zero recall, and calculator add and subtract. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operation of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine'sintended use and

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performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the electrical discharge machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 (OSHA) that are applicable to the machine itself.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

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3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28, for Federal Services and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03, for the English (US) system or ISO 54, for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

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3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The Electrical discharge machine shall consist of, but not be limited to, the following components: Base and column, saddle, table, ram or quill head assembly and dielectric coolant system.

3.4.1 Base and column. The base and column shall be designed to assure maximum rigidity and shall be stress relieved and normalized for dimensional stability. The base or a separate tank shall be used as a reservoir for the dielectric coolant. Doors or readily removable covers shall be provided in the base to permit servicing the dielectric reservoir and its components. Means shall be provided for leveling the machine.

3.4.2 Saddle. The saddle shall be mounted on the guideways provided on the base structure. Ways or guides shall be provided at the top of the saddle to incorporate a sliding table. The saddle shall have a hand-feed mechanism that will feed the saddle in a direction square with the movement of the table within the travel range specified in tables I or II. The hand-feed mechanism shall consist of a precision-feed screw and feed-screw nut, a hand wheel, and a graduated dial conforming to 3.3.7. When recirculating linear ball bushings, or rollers, on bar ways are used, each bar way shall be fully supported throughout its length and shall be hardened to not less than Rockwell C-60. Ball bushings, when utilized, shall be of the adjustable diameter type.

3.4.3 Table. The table shall be mounted on ways or guides on the saddle and shall travel square to the movement of the saddle. The table shall have a hand feed mechanism consisting of a precision leadscrew and leadscrew nut, a graduated dial conforming to 3.3.7 and a hand wheel. The feed mechanism shall be constructed to such precision as to allow positioning of the table to the accuracy specified in table III. When recirculating linear-ball bushings on bar ways are used, each bar shall be fully supported throughout its length and shall be hardened to not less than Rockwell C-60. Ball bushings, when utilized, shall be of the adjustable diameter type. The table shall have the number of T-slots specified in table I or II. T-slots shall conform to ANSI B5.1.

3.4.3.1 Table work tank. The table work tank shall be a part of, or attached to, the work table and shall be of a metal not adversely affected by the dielectric coolant. The dimensions of the work tank shall meet the requirements of table I or II. The front panel of the work tank shall be removable to facilitate installation, inspection and removal of the workpiece and the electrode. The removable front panel shall be designed to prevent leakage

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of the dielectric coolant when attached and secured in place on the work tank. The sealing material on the front panel shall not be adversely affected by the dielectric coolant. Means shall be provided for controlling the dielectric coolant level and automatically disconnecting electric power to the electrode if the coolant level falls below a predetermined level.

3.4.3.2 Table and saddle clamping devices. The saddle and table shall have an adequate number of clamps to hold the table or saddle in any pre-set position throughout their entire travel movement. Clamps shall be located in positions convenient to the operator. All clamps shall be designed so that the position of the clamped member is not disturbed during either engagement or disengagement of the clamp.

3.4.4 Ram or quill head assembly. The ram or quill head assembly shall be a casting or fabrication attached to the column and designed to meet the performance accuracy requirements in 3.6. The head assembly shall have scraped and flaked ways, recirculating ball bushings, pre-loaded roller bearings, or self-lubricating bearing pads to support the ram or quill and shall also contain an electro-hydraulic feed system. If a counterbalance system is required to balance the weight of the moving parts, it shall be completely housed in the head and column assembly.

3.4.4.1 Ram head. The ram head shall travel vertically, within the head slide to meet the requirements of table I and meet the alignment tolerances in table III. The ram shall have either dovetail, square, "V" type ways or a linear ball bushing and guide-rod way system. The ways, if of the dovetail, square or "V" type, shall have means for adjustment to compensate for wear and to maintain alignment accuracies of the ram. The rigidity of the ram shall conform to the requirements of table III.

3.4.4.1.1 Platen. A platen shall be mounted on the lower end of the ram and shall be electrically insulated from the ram assembly to prevent conduction of electrical current through ways or bearings. The platen shall have means for mounting the electrode. Such means shall consist of T-slots of the quantity and size specified in table I or II and means for mounting a collet chuck and collets when such operation is desired. T-slots shall conform to the requirements of ANSI B5.1.

3.4.4.2 Quill head. Class II machines shall be of the quill type designed for vertical movement in the workhead assembly and shall have a movement no less than the travel distance specified in table II. The travel of the quill shall be square with the work table surface within the tolerance specified in table III. Bearing support shall be at the top and bottom of the quill carrier to provide uniform support when the quill is fully extended. A rotating spindle shall be built in as an integral part or attached to the quill and shall be powered by a variable speed motor having a range of 0 to

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300 RPM. Provisions shall be provided to lock the spindle to prevent rotation when not in use. The end of the quill shall have provisions to allow dielectric coolant to be pumped through drilled passages in the electrode. The electrode holder shall be electrically insulated from the quill assembly to prevent conduction of electrical current through the ways or bearings.

3.4.4.3 Ram and quill feed system. The ram and quill feed systems shall be driven by either a hydraulic cylinder, or a hydraulic motor with a gear train, and a precision leadscrew. The feed system shall have all necessary controls to assure proper positioning to the electrode in relation to the workpiece mounted on the table. The control system shall be electrically integrated with the power supply and shall maintain the pre-set conditions of spark gap, metal removal rate, and surface finish. Means shall be provided to automatically shut off the servo feed mechanism to minimize drift when machining power is turned off. It shall be possible to manually operate the feed system for setup purposes and shall also be possible to control the movement of the ram or quill with machining power either on or off.

3.4.4.4 Ram or quill stop and indicator. A ram or quill stop and a dial type indicator shall be provided. Both the stop and dial indicator shall be graduated in increments of 0.001 inch or less, and both shall be capable of use at any position within the stroke of the ram or quill. The dial indicator shall indicate the downward movement of the ram or quill and the stop shall stop the ram or quill at any pre-set position.

3.4.5 Dielectric coolant system. The dielectric coolant system shall consist of a reservoir in the base of the machine or an auxiliary tank, as applicable, a fast filling and emptying system, a pumping system to supply coolant to the electrode and workpiece, a filtering system, necessary piping, valves, and electrical controls.

3.4.5.1 Fast filling and emptying system. A fast filling and emptying system shall be provided for filling or emptying the table work tank to allow rapid loading of the workpiece and adjustment or inspection of the electrode and workpiece. The system shall employ either a pressurized reservoir operating from an 80 to 100 PSI shop compressed air system (source of air not required to be furnished) or a high capacity pump (which shall be furnished). The system shall be capable of delivering coolant to, and removing coolant from, the work tank at a rate of not less than 55 gallons per minute.

3.4.5.2 Dielectric coolant pump system. The pressure pump supplying dielectric coolant to the electrode and workpiece shall be of a type not adversely affected by the coolant or abrasive particles. The pump shall be powered by an electric motor of a continuous rating to meet the requirements of table I or II. The pump shall be capable of providing either pressure or

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vacuum at the electrode or workpiece. A means shall be provided for monitoring the coolant pressure at the pump outlet. Pressure relief valves shall be provided when required to prevent build-up of excessive pressure.

3.4.5.3 Filter system. A coolant system shall have a replaceable or washable element type filter. Pressure gauges shall be provided to indicate the pressure drop across the filter to enable the operator to determine when the element needs cleaning or replacing. The filter shall be located in an accessible position to facilitate servicing.

3.4.5.4 Piping, hoses, valves. All piping, hoses and valves shall be of adequate size to minimize pressure drop in the coolant system. All hoses shall be fabricated of a material not adversely affected by the coolant and shall have a burst pressure rating not less than four times the maximum pressure that can be developed in the system. Hoses shall be of adequate length so that they are not stretched or strained when the parts to which they are connected are in their extreme positions and shall be routed and attached in a manner that will prevent rubbing and chafing. The piping system shall have all necessary regulating valves, relief valves, and shut-off valves required to regulate the flow of coolant to the various circuits of the coolant system. Adequate couplings and unions shall be provided to allow a component to be removed without disassembly of the remainder of the system.

3.4.6 Electrical output characteristics. Unless otherwise specified (see 6.2), the power supply unit shall have two sets of output terminals and two sets of controls capable of supplying 50 percent of the maximum current to each of two electrical discharge machines in dual operations, or both sections shall be used together to supply the maximum output current to one machine in a single operation.

3.4.6.1 Capabilities. Necessary controls and circuitry shall be furnished to provide control of the metal removal rate, overcut, surface finish, and electrode no wear. The machines shall have the capability of removing 0.05 cubic inch of metal per hour per ampere when machining steel with a graphite electrode.

a. Units having current output capacities of 20 to 150 amperes shall be capable of controlling the overcut over a range of 0.0005 to 0.004 inch per side and producing a surface finish of 10 microinches.

b. Units having current output capacities of more than 150 amperes shall be capable of controlling the overcut over a range of .003 to 0.15 inch per side and producing a surface finish of 100 microinches.

All units shall have the circuitry necessary to prevent the erosion of the electrode when adjusted to the no wear mode. All power supply units shall be

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integrated with the electrode feed controls such that the proper electrode gap spacing is automatically maintained for any selected combination of output pulse frequency, gap voltage, and current.

3.4.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.8 Motors. Motors shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operated on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1.

3.4.9 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.4.10 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of machining through holes, blind holes, shaped cavities, surface contours and trepanning of hardened steels, exotic metals and high temperature alloys.

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table III.

3.8 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be attached in such a way to provide for adjustment to reflect maximum light on the work area. If a built-in light or lighting system is provided to illuminate the work area, the adjustable light described above shall not be required.

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3.9 Standard equipment. All standard equipment normally required to properly operate the Electrical Discharge Machine such as special tools shall be furnished.

3.9.1 Optional equipment. When specified (see 6.2), the following optional equipment shall be furnished:

a. Collet chuck and collet set 1/16 - 1/4 inch consisting of (7) collets in 1/32-inch increments. (Adapters, if required, to fit chuck to platen or quill shall be furnished).

b. Collet chuck and collet set 1/4 - 3/4 inch consisting of (17) collets in 1/32-inch increments. (Adapters, if required, to fit chuck to platen or quill shall be furnished).

c. Tapping attachment (ram type machine).

d. Vibrating electrode holder.

e. Ram or quill cycling unit.

f. Adjustable V-block holder.

g. Optical measuring device for table and saddle.

h. Pickup, locating, and tramming attachment complete with dial indicator.

i. Other required equipment not described above.

3.10 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication plate. Unless otherwise specified (see 2.1), a corrosion-resistant lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity (Military or Federal specification number)

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine and contain the information listed

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below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Hour meter. Unless otherwise specified (see 6.2.1), the Electrical Discharge Machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.13 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423, Contract Data Requirements List. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the electrical discharge machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for compliance. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification

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shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspections shall consist of the examination in 4.6 and all test in 4.7. Failure of the first article tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspections shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the test, or the inspections shall be cause for rejection.

4.6 Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of table III (see 3.7.).

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4.7.3 Performance tests. Machines with 20, 30, 40, 60, 80, 100, and 150 amp power supply units shall be subjected to all of the tests described in 4.7.4, 4.7.5 and 4.7.6 through 4.7.9. Machines with 200, 300, 400 and 500 amp power supply units shall be subjected to the tests described in 4.7.4, 4.7.5(b), 4.7.7 and 4.7.9.

4.7.4 Maximum metal removal. Using an oil-hardening tool steel workpiece and a graphite electrode having a diameter not less than 1.000 inch, the machine shall demonstrate that it is capable of eroding 0.05 cubic inch of metal per hour per ampere. For the purpose of this test, the volume of the core created by the coolant passage, shall be used in calculating the removal rate.

4.7.5 Smoothest surface finish. Using an oil-hardening tool steel workpiece and a copper electrode having a diameter not less than 1.0000 inch and a coolant passage not larger than 1/8 inch and with the controls adjusted for smoothest surface finish, the machine shall erode a spot face and produce a surface finish conforming to the following requirements:

a. Machines having a 20, 30, 40, 60, 80, 100 and 150 amp power supply shall produce a 10 microinch finish.

b. Machines having a 200, 300, 400, and 500 amp power supply shall produce a 100 microinch finish.

The above requirements are without regard to metal removal rate or amount of overcut.

4.7.6 Overcut cut. A through-hole not larger than 1.0000 inch outside diameter shall be eroded through an oil-hardening tool steel workpiece not less than 0.375-inch thick. A metallic electrode having an outside diameter of 1.1000 $\pm$0.0001 inch and a coolant passage shall be used to enlarge the hole with the controls adjusted for minimum overcut. The diameter of the finished hole, measured at the entering side, shall be measured. The overcut shall be calculated and shall be not more than 0.0005 inch per side.

4.7.7 Structural rigidity test. Using a steel workpiece of adequate thickness to allow precise measurements after completion of the test, the machine shall erode a through-hole using an electrode having a cross section area not less than 20-square inches with the coolant pressure adjusted to not less than 30 psi. The perpendicularity of the hole sides to the supporting surface of the workpiece shall be measured. The sides of the hole shall be perpendicular to the supporting surface within 0.0005 inch per foot in all directions.

4.7.8 Small hole test. Using an oil-hardening tool steel workpiece not less than 0.030-inch thick, and a metallic electrode having a diameter not

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larger than 0.063 inch and having no coolant hole, the machine shall be capable of eroding three through-holes without deformation or breakage of the electrode.

4.7.9 Electrode erosion control test. Using an oil-hardening tool steel workpiece not less than 0.375-inch thick and a graphite electrode having a diameter of 1.0000 inch and with the controls adjusted for no wear on the electrode and for a metal removal rate not less than 60 percent of the maximum rate, the machine shall erode a through-hole. The diameter of the electrode shall be measured throughout its active length and shall be not less than its original dimensions.

4.7.10 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058 as specified.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended to be used for die sinking operations, and reproducing complex shapes and cavities to a high degree of accuracy in a wide range of materials, particularly difficult-to-machine metals.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and data of this specification.

b. Class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If electrical discharge machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Digital readout system, specify axis of travel required and any special requirements if required (see 3.2.1.2).

g. Specify each axis that requires and electronic display (see 3.2.1.2).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

j. Electrical output characteristics (see 3.4.6).

k. Electrical system and input power,if different (see 3.4.7).

l. Motor enclosure, if different (see 3.4.8).

m. Size and capacity, if different (see 3.5).

n. Worklight, if required (see 3.8).

o. Specify and fully describe optional equipment, if required (see 3.9.1).

p. Specify applicable repair parts, if required (see 3.10).

q. Lubrication chart or plate, if different (see 3.11.1).

r. Nameplate, if different (see 3.11.2).

s. Hour meter, if different (see 3.12).

t. Inspection conditions, if different (see 4.3).

u. First article inspection, if different (see 4.4).

v. Quality conformance inspection, if different (see 4.5).

w. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

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6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (keyword listing).

Coolant system, dielectric
Feed system, ram and quill
Head assembly, ram or quill head

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TABLE I. Sizes and capacities (see 3.5).

Characteristic	Class 1 machines				
	Size 1	Size 2	Size 3	Size 4	Size 5
Ram workhead					
Ram travel, inches	8	8	12	12	12
Throat distance, centerline of platen to face of column, inches	10-1/2	15	16	22	17
Platen size, inches	6 x 8	7 x 8	7 x 10	9 x 12	10 x 14
Platen to top of table, inches, range	9 to 17	9 to 14	10 to 22	10 to 22	10 to 30
T-slots, quantity	1	2	2	2	3
Table working surface, inches	20 x 9	20 x 12	20 x 12	36 x 12	40 x 24
Tank size (inside dimensions), inches	24 x 16 x 12	30 x 20 x 12	24 x 20 x 13	41 x 20 x 13	50 x 20 x 26
Table T-slots, quantity	1	3	3	3	3
Table longitudinal movement, inches	10	12	12	24	8
Table cross movement, inches	6	10	12	12	8
Motor sizes					
Coolant pressure pump, HP	1/2	1/2	1/2	1/2	1/2
Coolant fast-fill pump, HP	1/3 or air	1/3 or air	1/3 or air	1/3 or air	1/3 or air
Hydraulic pump, HP	1/2	1/2	3/4	1/2	1/2
Dielectric reservoir					
Capacity, gallons	55	80	110	110	165

NOTE: Machine sizes and capacities shall be less than the stated requirements for the size ordered nor greater than the requirements of the next larger size as shown in table I. When the largest size machine is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Sizes and capacities (see 3.5).

Characteristic	Class 2	
	Size 1	Size 2
Quill workhead		
Quill travel, inches	8	8
Throat distance, centerline of quill to face of column, inches	15-1/2	16-1/4
End of quill to top of table, inches, range	3 to 18	6 to 22
Machine table and tank		
Table working surface, inches	26 x 14	29 to 14
Tank size (inside dimensions), inches	42 x 22 x 12	48 x 22 x 14
Table T-slots, quantity	3	3
Table longitudinal movement, inches	20	20
Table cross movement, inches	11	12
Motor sizes		
Press pump, HP	1/2	1/2
Fill-pump, HP	1/2 or air	1/2 or air
Hydraulic pump, HP	1/2	1/2
Dielectric reservoir		
Capacity, gallons	1-5	70

NOTE: Machines sizes and capacities shall be not less than the stated requirements for the size ordered nor greater than the requirements of the next larger size as shown in table I. When the largest size machine is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE III. Alignment tolerances.

Characteristic	Tolerance (not greater than)
Table level in longitudinal and transverse directions	0.005" per ft.
Table flat in longitudinal and transverse directions	0.005" per ft. (concave only)
Table travel square with saddle	0.0008" per ft.
Table travel straight in longitudinal and transverse directions	0.005" per ft.
Rise and fall of table, while traversing longitudinally and transversely, full stroke	0.0005" per ft.
Center T-slot parallel with longtiduinal travel of table	0.0005" per ft.
Positioning error of table and saddle	±0.0005" per 6.000"
Electrode holder axis (taper or collet) square with table surface	0.001" per ft.
Platen surface parallel with table surface	0.001" per ft.
Rigidity of ram in extreme bottom position, and with a force of 50 pounds applied alternately from side to side, and from front to rear, deflection of extreme bottom or ram, in either direction shall not exceed:	.0025" total movement (zero the indicator with 50# force applied in one direction: read indicator when force is applied in opposite direction)
Rigidity of quill with quill in extreme bottom position, and with a force of 10 pounds applied alternately from side-to-side and from front-to-rear, deflection of extreme bottom of quill, in either direction shall not exceed:	.006" total movement (zero the indicator with 10# force applied in one direction: read indicator when force is applied in opposite direction)

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Custodians:
Navy - SH
Air Force - 99

Review activities:
Army - CR
Navy - AS, MC
Air Force - 84
DLA - GS

User activities:
Army - AR
Navy - OS

Preparing activity:
DLA - IP
Project (3410-0039)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-80023D	Electrical Discharge Machine, Vertical, Ram and Quill Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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